

SEMESTER-VI

UEC702: MICROCONTROLLER AND EMBEDDED SYSTEMS				
	L	T	P	Cr
	3	0	2	4.0
Course Objective: Embedded systems are pervasive in all areas of society, and as such, knowledge of how to design them is a vital skill for all electrical engineers. The objective of this course is to equip students with the necessary fundamental knowledge and skills that enable them to design basic embedded systems, where a microprocessor or microcontroller is the central element.				
Syllabus Introduction to Processor Design General purpose processor architecture and organization, Von-Neuman and Harvard architectures, Processor design trade-offs, CISC and RISC architectures, Advantages of RISC architecture, Processor cores: soft and hard. Microcontroller: Introduction to Microcontrollers, Evolution, Microprocessors vs. Microcontrollers, MCS-51 Family Overview, Important Features, 8051 Pin Functions, Architecture, Addressing Modes, Instruction Set, Instruction Types. Programming: Assembly Programming. Timer Registers, Timer Modes, Overflow Flags, Clocking Sources, Timer Counter Interrupts, Baud Rate Generation. Serial Port Register, Modes of Operation, Initialization, Accessing, multiprocessor Communications, Serial Port Baud Rate, Interrupt Organization, Processing Interrupts, Serial Port Interrupts, External Interrupts, Interrupt Service Routines. ARM Introduction and Pipeline structures ARM processor architecture, Endianness, Processor core VS CPU core, ARM7TDMI Interface signals, Memory Interface, Bus Cycle types, Register set, Operational Modes. Instruction Format, ARM Core Data Flow Model, ARM 3 stage Pipeline, ARM family attribute comparison. ARM 5 stage Pipeline, Pipeline Hazards, Data forwarding ARM7TDMI assembly instructions and modes ARM ISA and Processor Variants, Different Types of Instructions, ARM Instruction set, condition codes, Load-Store instructions, Data processing instructions, Shift Operations, Arithmetic, Logical instructions, Addressing modes. Control Flow Instructions, Interrupts and Exceptions, software Interrupt Instruction, Assembly Language Programming, Thumb state, Thumb Programmers model, Thumb Applications, ARM coprocessor interface and Instructions, Coprocessor Instructions, ARM tools and Peripherals ARM Development Environment, Arm Procedure Call Standard (APCS), Example C/C++ programs, Embedded software development, Image structure, linker inputs and outputs, ARM systems with MPU, memory Protection Unit (MPU). Physical Vs Virtual Memory, Paging, Segmentation. AMBA Overview, Typical AMBA Based Microcontroller, DMA, Peripherals, Programming Peripherals in ARM.				
Course Learning Outcomes (CLOs) The students will be able to: 1. understand need of Microcontroller, its architecture and assembly programming. 2. understand ARM processor architecture, instructions and their formats. 3. write ARM processor-based assembly language programs.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

4. interface coprocessor, AMBA bus architecture, various HW peripherals in embedded systems and how memory mapping can be done.
5. understand issues to be handled in any processor software tool chain for embedded system development especially using C/C++.

Text Books

1. The 8051 Microcontroller Mckenzie, Scott,, PHIs, 5th ed. (1995)
2. “Computer Organization and Embedded Systems, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, Naraig Manjikian, McGraw Hill, Sixth Edition, 2012.
3. “ARM System-on-Chip Architecture, Steve Furber, PEARSON, Second Edition, 2013.

Reference Books

1. The 8051 Microcontroller: Architecture, Programming, and Application, Ayala, Kenneth J., 2nd ed,2008
2. “ARM: Assembly Language Programming”, Stephen Welsh, Peter Knaggs, Bourne Mouth University Publication, 2003.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	40
3	Sessionals (Assignments/Projects/ Lab Evaluations)	35

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